nature photonics

Article

https://doi.org/10.1038/s41566-026-01880-9

Experimental memory control in continuous-variable optical quantum reservoir computing

Received: 26 June 2025

Accepted: 22 February 2026

Published online: 17 March 2026

Check for updates

Iris Paparelle[1,2] ✉, Johan Henaff[1], Jorge García-Beni[2], Émilie Gillet[1], Daniel Montesinos[2], Gian Luca Giorgi[2], Miguel C. Soriano[2], Roberta Zambrini[2] & Valentina Parigi[1] ✉

Forecasting complex processes requires efficient learning from temporal data. Reservoir computing platforms enable such learning with minimal training cost. Quantum reservoir computing (QRC) extends this framework into the quantum domain, offering promising capabilities for online, quantum-enhanced machine learning tailored to temporal tasks. As in the classical case, photonics provides a natural platform for QRC. However, implementing native memory capabilities in practical photonic quantum systems remains a major challenge. Here we demonstrate a photonic QRC platform based on deterministically generated multimode squeezed states, exploiting spectral and temporal multiplexing in a continuous-variable setting with controllable fading memory. Data is encoded via programmable pump phase shaping in an optical parametric process and retrieved through mode-selective homodyne detection. Real-time memory is implemented through feedback via electro-optic modulation, and expressivity is boosted via spatial multiplexing. This architecture enables nonlinear temporal tasks, including parity check at different delays and chaotic signal forecasting. All results are supported by a high-fidelity Digital Twin. Leveraging the entangled multimode structure enhances expressivity and memory capacity, establishing a scalable continuous-variable photonic platform for quantum-enhanced information processing.

Reservoir computing is a machine learning neuro-inspired paradigm that exploits the dynamics of an artificial network with minimal learning requirements. Unlike conventional neural networks, it avoids the computational cost of tuning internal weights during training; instead, only the output layer is trained, typically via linear regression[1,2]. The untrained network, known as the reservoir, can be implemented as a physical process.

Although the term reservoir computing is now broadly used for tasks involving an untrained network (that is, a reservoir), its most distinctive feature is the ability to learn from temporal series (for example, predicting chaotic time series) by exploiting the memory properties of the system used as a reservoir. This temporal processing capability enables online learning and extends beyond static tasks such as classification traditionally addressed by architectures such as extreme learning machines[3,4].

Quantum reservoir computing (QRC) extends the concept of physical reservoirs into the quantum regime and has recently been extensively explored across several platforms[5–8] revealing both novel

[1]Laboratoire Kastler Brossel, Sorbonne Université, ENS-Université PSL, CNRS, Collège de France, Paris, France. [2]Instituto de Física Interdisciplinar y Sistemas Complejos (IFISC), UIB-CSIC UIB Campus, Palma de Mallorca, Spain. ✉e-mail: iris.paparelle@lkb.upmc.fr; valentina.parigi@lkb.upmc.fr

capabilities[9–15] and new challenges[16,17]. Photonic platforms are particularly appealing as they inherit the capabilities of their classical counterparts[18,19], offer many exploitable degrees of freedom, possess high integration potential and, when information is encoded in quadrature continuous variables, can operate at room temperature using coherent detection for online monitoring[20].

Photonic systems have recently been tested in quantum machine learning tasks[21,22]. In the context of QRC, following different proposals in either static[23,24] and temporal tasks[10,20,25], some pioneering experimental results have been demonstrated on quantum optical platforms; these have primarily focused on classification tasks[15,26,27] or classically assisted strategies[28]. Only recently have efforts been directed at incorporating memory capabilities into general quantum platforms such as superconducting circuits[29–31], and into a memristor-based photonic platform[32].

Here we present an experimental realization of memory-tailored QRC using a reconfigurable, multimode continuous-variable (CV) photonic platform. Memory is engineered through feedback conditioned on the history of the quantum system, enabling online temporal learning. Our approach exploits deterministically generated squeezed and reconfigurable entangled quantum states, multiplexed across both spectral and temporal modes[33–35] along with reconfigurable mode-selective homodyne detection. The experimental realization of temporal learning tasks via CV builds on theoretical results that demonstrate the potential of such systems in QRC[13,20,25,36]. The experimental set-up supports controllable memory demonstrating a performance boost in online learning and does not require post-processing with a classical neural network platform.

Data is encoded either via modulation of the global phase or of the complex spectral shape of the pump pulses driving the nonlinear optical process. Global phase modulation imprints information into the angle of the squeezing ellipses, tuning quantum correlations in the measurement basis. Spectral shaping directly modifies quantum correlations among output modes by altering the nonlinear interaction itself.

Moreover, we construct a Digital Twin of the experiment by integrating accurate models of the quantum and nonlinear optical processes, along with the CV dynamics of the reservoir. Once calibrated with a noise model, this simulation framework accurately reproduces experimental results and enables predictive exploration of the platform's capabilities.

We experimentally perform temporal tasks enabled by memory, using global phase encoding. Memory is achieved via a feedback signal applied to the electro-optic phase modulator (EOM)[37] that controls the pump phase; the feedback is derived from observables measured via mode-selective homodyne detection of the quantum reservoir's multispectral components. For short-memory tasks, we implement real-time feedback control, while longer-delay target functions are experimentally realized using a spatial multiplexing strategy, that is, by running sequential experiments that act as spatially distributed copies of the quantum reservoir platform.

Finally, we demonstrate that fully exploiting the entangled multimode nature of the system yields better scaling of reservoir expressivity when compared with the use of multiple independent copies of the experiment (spatial multiplexing) and enhance the memory in temporal tasks.

This work demonstrates online temporal processing enhanced by controllable memory using a CV entangled reservoir. Gaussian states and homodyne detection have been theoretically shown to increase information processing capacity compared with classical encodings[25], allowing saturation of a polynomial advantage as the reservoir size grows[20]. This regime is experimentally explored for the first time in this article, with potential implications for practical noisy intermediate-scale quantum implementations. Furthermore, our architecture is compatible with measurement-based approaches using cluster states[13] and with the integration of non-Gaussian resources[15,38].

Defining quantum advantage in QRC or quantum machine learning remains an open challenge[39]. Nevertheless, CV platforms enhanced with non-Gaussian resources are central to fault-tolerant photonic quantum computing[40]. Our demonstration of memory-enhanced online temporal tasks enables scalable QRC using only deterministic Gaussian-state generation and room-temperature homodyne detection, providing a basis for exploring quantum advantage with non-Gaussian resources[41–44].

Results

The quantum resource

The core quantum resource is a multimode optical network generated via parametric down-conversion (PDC) in a nonlinear waveguide pumped by a pulsed laser, as shown as in Fig. 1. The interaction Hamiltonian describing the generation of signal and idler fields of wavelengths λ_S, λ_I reads

$$\hat{H} \propto \int d\lambda_S d\lambda_I J(\lambda_S, \lambda_I)\, \hat{a}^\dagger(\lambda_S)\hat{b}^\dagger(\lambda_I) + \text{h.c.}, \tag{1}$$

where $J(\lambda_S, \lambda_I)$, the so-called joint spectral amplitude (JSA), can be written as[45–47]

$$J(\lambda_S, \lambda_I) = p(\lambda_S, \lambda_I) \cdot \phi(\lambda_S, \lambda_I). \tag{2}$$

The term $p(\lambda_S, \lambda_I)$ is the complex pump spectral shape, while $\phi(\lambda_S, \lambda_I)$ is the phase-matching function, determined by the characteristics of the nonlinear medium. The multimode nature of the generated quantum system can be retrieved by performing a Schmidt decomposition of the JSA $J(\lambda_S, \lambda_I) = \sum_k r_k h_k(\lambda_S) g_k(\lambda_I)$, where $h_k(\lambda_S)$ and $g_k(\lambda_I)$ are two sets of orthonormal modes, spectral profiles of signal and idler channels, and r_k are called the Schmidt coefficients. In case of degenerate signal and idler fields, as in our experimental set-up, $h_k = g_k$; and the Hamiltonian in equation (1) can be written as $\hat{H} \propto \sum_k r_k \hat{A}_k^{\dagger 2} + \text{h.c.}$, with $\hat{A}_k^\dagger$ broadband creation operators associated to the modes h_k. Acting on the vacuum, this Hamiltonian generates a tensor product of single-mode squeezed vacuum states, one in each mode $h_k(\lambda_S)$, also called supermodes. The used set-up can generate up to 40 Schmidt modes approximated by Hermite–Gauss polynomials as shown in Fig. 1.

The generated quantum state is Gaussian and fully characterized by the quadrature covariance matrix $\boldsymbol{\sigma}$, measured via homodyne detection. Mode-selective homodyne detection enables probing in arbitrary mode bases. In the supermode basis, $\boldsymbol{\sigma}$ is diagonal; in any other basis $\xi_j(\lambda_S)$ related by a unitary $\boldsymbol{U}$, it transforms as $\boldsymbol{\sigma}' = \boldsymbol{S}_U^T \boldsymbol{\sigma} \boldsymbol{S}_U$, where $\boldsymbol{S}_U$ is the symplectic transformation induced by $\boldsymbol{U}$ (Supplementary Section B). The matrix $\boldsymbol{\sigma}'$ is non-diagonal and the off-diagonal terms are signatures of entanglement correlations between the measured modes. Measuring on such basis allows to experimentally access entanglement correlations for quantum protocols[35,48]. The resource can then be pictured via an entangled network as shown in Fig. 2.

The phase-matching function term of the JSA in equation (2) is fixed and determined by the material and waveguide properties, and only the pump-envelope function can be modified in a reconfigurable way via mode-shaping techniques. This modifies the entanglement correlation between nodes as shown in Fig. 2. Therefore, we can explicitly write the dependence of the covariance matrix $\boldsymbol{\sigma}'$ on the pump p and the chosen basis U as $\boldsymbol{\sigma}' = \mathbf{S}_\mathbf{U}^T \boldsymbol{\sigma}(\mathbf{H}(\mathbf{p}))\, \mathbf{S}_\mathbf{U}$.

In the following, we exploit the control on the pump-envelope function to encode information in the quantum system in the configuration of a fixed measurement basis U. The expectation value of general observables $O_m = \langle \Delta\hat{x}_i \Delta\hat{x}_j \rangle$ that will be used in the reservoir task can be retrieved from the covariance matrix $\boldsymbol{\sigma}'(U, p)$. Given the pump shape, we have accurate numerical models of the process from the nonlinear interaction to the observables[45–47], setting a Digital Twin of the experimental set-up as fully explained in Supplementary Section A.

https://doi.org/10.1038/s41566-026-01880-9

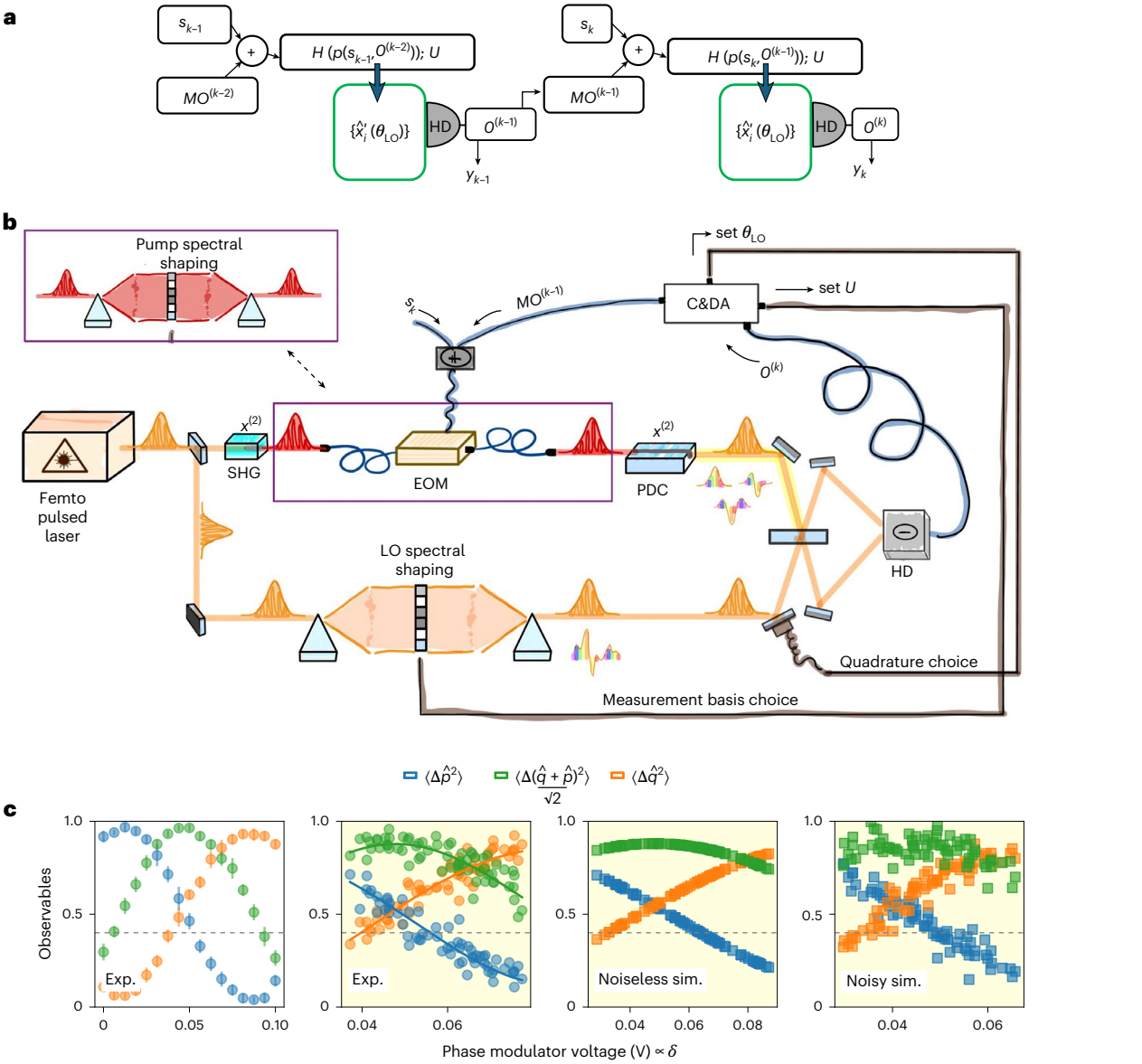


Fig. 1 | QRC: protocol, set-up and observable dynamics. a, Quantum reservoir protocol. At each timestep k, the input s_k is added to a feedback signal derived from the previous observables $O^{(k-1)}$. This sets the pump of the parametric process, determining the operators $\hat{x}'_i$ and their quantum correlations. The task output y_k is obtained from the observables. **b**, Experimental set-up. Information is encoded in the global phase of the pump, generated via second harmonic generation (SHG) and modulated by an EOM. The more general encoding involves phase shaping of the pump spectrum. The multimode entangled state from PDC, and thus its observables, is tailored by this encoding. The control and data acquisition (C&DA) module includes electronics to set the measurement basis (U) via local oscillator shaping in homodyne detection (HD), measure the observables O_m and generate the feedback. Phase locking is not shown. **c**, Nonlinear dependence of observables with the EOM voltage (proportional to the encoding phase δ, which is a linear combination of s_k and $O^{(k-1)}$). Left: experimental data (dots) in low- and average-noise regimes. The low-noise regime (first panel) is obtained by repeating the measurement ten times; the data are shown as mean values ± s.d. All the other panels present one-sample independent measurements and simulations to highlight the impact of the noise and therefore present no error bars. Right: simulated data (squares) using the Digital Twin model according to the experimental noise evaluation. The white panel spans a voltage range sufficient for observables to vary from 0 to 1 (phase difference slightly exceeding $\pi/2$). Yellow panels show observables recorded during training on a memory task.

Information encoding and memory control of the quantum reservoir

Tuning the pump phase profile, the pump-envelope function can be written as

$$p(\lambda_P, \boldsymbol{\delta}) = \sum_{i=1}^{N} f_i(\lambda_P) e^{i\delta_i}, \tag{3}$$

where the $f_i(\lambda_P)$ are a set of N pump modes and the $\boldsymbol{\delta} = (\delta_i)_{i=1}^{N} \in \mathbb{R}^N$ are tunable phase parameters that allow precise shaping of the JSA to easily encode information. The pump input modes f_i that we consider are N frequency windows, delimiting a portion of our Gaussian pump profile $f(\lambda_P)$, that we can precisely control with a spatial light modulator, for example. In the simple case of $N = 1$, when a unique global phase parameter is considered, an EOM is used as shown in Fig. 1. To perform the measurement, a set of n modes $\xi_j(\lambda_S)$ is selected by the basis choice U via the local oscillator shaping in homodyne. The tailoring of N pump spectral components and the choice of the measurement basis composed by n elements are shown in the upper part of Fig. 2 (more details in Supplementary Section B).

https://doi.org/10.1038/s41566-026-01880-9

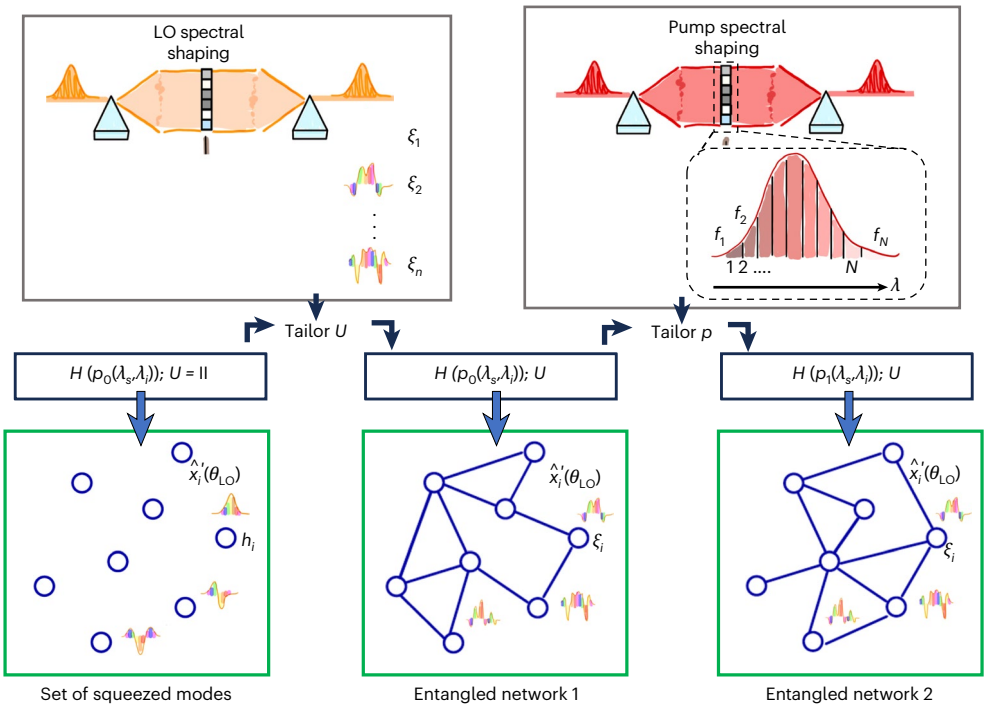


Fig. 2 | Controllability of the resource by tailoring the local oscillator and the pump spectral shapes. The nodes of the network are different spectro-temporal modes of the field. The set of modes are defined by the basis choice U. For convention when $U = I$, the supermode basis $\{h_i\}$ is chosen, and the modes are independently squeezed. In all the other mode bases $\{\xi_i\}$, the modes show entanglement correlations represented as links in a network. The general mode quadrature is $\hat{x}'_i(\theta_{LO}) = \cos(\theta_{LO})\hat{q}'_i + \sin(\theta_{LO})\hat{p}'_i$ that corresponds to $\hat{q}'_i$ or $\hat{p}'_i$ if the local oscillator phase is set to be $\theta_{LO} = 0$ or $\theta_{LO} = \pi/2$. The correlations can be controlled by tailoring the pump of the process that modifies the Hamiltonian.

Our system is endowed with memory through a real-time feedback mechanism: the results of previous measurements are fed back into the system to influence future states. Specifically, if we consider a discrete-time sequence of inputs $\{s_k\}$, where k denotes the timestep, the pump parameters are updated according to

$$\delta_i^{(k)} = \alpha_i s_k + \beta_i + \sum_m M_{im} O_m^{(k-1)}. \tag{4}$$

Here $i \in 1, \ldots, N$, α_i and β_i are real-valued coefficients, and M_{im} is a feedback mask (a real matrix) that determines how previous measurement outcomes influence the current pump parameters. This determines the fading memory of the QRC. To summarize, N is the number of different input components that we encode in a single instance by manipulating several components of the pump spectrum: $N > 1$ can encode multidimensional inputs as well as enrich the encoding of a scalar input, repeated across the pump spectrum with different parameters $(\alpha_i, \beta_i, M_{im})$. The quantities $O_m^{(k-1)} = \langle \Delta\hat{x}_i \Delta\hat{x}_j \rangle^{(k-1)}$ are the values of the mth observable measured at timestep $k-1$, and are derived from a subset of elements of the covariance matrix $(i,j \in 1, \ldots, 2n)$. These are measured by accumulating data on the homodyne signal. The number n of measured modes determines then the size of the output layer of our reservoir. The reservoir computing is described by

$$\begin{cases} O_m^{(k)} \propto \sigma'^{(k)}_{ij} | \boldsymbol{\sigma}'^{(k)} = \mathbf{S}_\mathbf{U}^T \boldsymbol{\sigma}(\mathbf{H}(\mathbf{p}(\boldsymbol{s}_\mathbf{k}, \mathbf{O}^{(\mathbf{k-1})}))) \mathbf{S}_\mathbf{U} \\ y_k = \mathbf{w}^T \mathbf{O}^{(k)} + b \end{cases}. \tag{5}$$

The reservoir is trained by adjusting the matrix $\mathbf{w}$ and parameter b to minimize the error in the chosen learning task. The scheme of the quantum reservoir with feedback is shown in the top part of Fig. 1. While we here focus on the case of feedback restricted to the last output at each cycle, the fading memory can be extended re-injecting a longer combination of past signals.

QRC tasks

We evaluate the performance of our system across a range of benchmark tasks considering two input encoding: the simple, experimentally implemented scheme with $N = 1$, and the more general encoding with $N > 1$. We begin with the XOR task, which assesses the system's nonlinearity and short-term memory while requiring minimal resources. We then consider the memory task to characterize the system's fading memory property. Finally, we turn to more complex benchmarks, including the double-scroll and parity check tasks, which challenge the system's scalability and capacity to handle richer, higher-order nonlinear dynamics.

Experimental control of memory via global phase. The phase encoding via the EOM, as shown in Fig. 1, corresponds to $N = 1$, and the voltage applied changes linearly the pump phase δ:

$$p(\lambda_P, \delta) = f(\lambda_P) e^{i\delta}, \tag{6}$$

where $f(\lambda_P)$ is the natural Gaussian profile of the pump. As explained in Methods, with this type of encoding, the expressivity of the observables does not increase with the number of measured modes n; therefore, we use $n = 1$ as well. This means that we measure only the first supermode h_1 and we can use as observables $\{O_1, O_2, O_3\}$ the set $\{\langle \Delta\hat{q}^2 \rangle, \langle \Delta\hat{p}^2 \rangle, \langle \Delta\hat{q}\Delta\hat{p} \rangle \sim \langle \Delta(\frac{\hat{q}+\hat{p}}{\sqrt{2}})^2 \rangle\}$. These three observables are sufficient for implementing several tasks, as they express three different nonlinear dependencies with respect to δ as shown in the lower part of Fig. 1. The first two panels show the three behaviours experimentally, in the case of a low-noise realization and a more noisy one, which is representative of the average noise in measurements.

https://doi.org/10.1038/s41566-026-01880-9

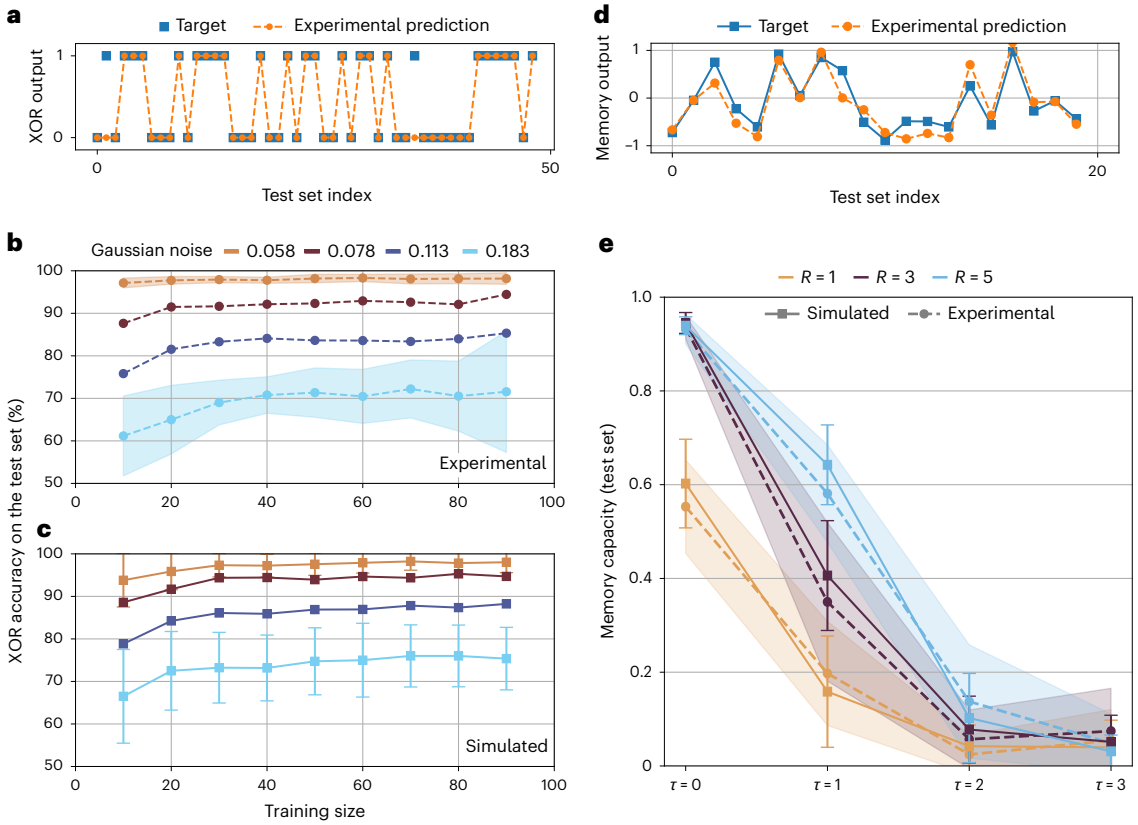


Fig. 3 | Experimental and simulated performance via global phase encoding. XOR task (experimental and simulated). **a**, Example of binary target (blue) and predicted (orange) outputs for the test set, with a test accuracy of 95.9% (correct predictions over total). Training and test set sizes are 99 and 49, respectively. **b**,**c**, Test accuracy versus training size: experimental data (dots) (**b**) and simulated data (squares) (**c**) under varying noise levels (light brown, brown, blue and light blue); these noise levels are determined by calculating the standard deviation around the expected behaviour of the observables (see the bottom of Fig. 1). Experimental error bars (shaded) represent mean ± s.d. over $n = 50$ independent reshufflings of the dataset (technical replicates). Simulated data points represent mean ± s.d. over $n = 100$ independent simulation runs (different random seeds; technical replicates). Error bars are shown only for the first and last noise levels for clarity. Larger training sets lead to smaller test sets, increasing variability owing to single misclassifications. Memory task (experimental and simulated). **d**, Continuous target (blue) and predicted (orange) output for a test set using 5 reservoirs with delay $\tau = 1$ (capacity = 0.81). **e**, Capacities versus delay τ: experimental (dots) and simulated (squares) results for different reservoir numbers (light brown, dark purple and light blue). Experimental error bars represent mean ± s.d. over $n = 100$ reshufflings (technical replicates), and simulated error bars represent mean ± s.d. over $n = 30$ simulation runs (technical replicates). In both cases, simulations use a realistic experimental noise (for the memory task, we had a Gaussian noise of 0.057) model via the Digital Twin.

Assuming a Gaussian noise model, the typical variance of observables is recovered via a least square method. The result is used in the Digital Twin to simulate the expected behaviour as shown in the last two panels.

The unique non-zero term M that we use for the feedback mask is the one corresponding to $\langle \Delta\hat{q}^2 \rangle^{(k-1)}$. At each timestep k, we set the global phase depending on the input s_k and the feedback as

$$\delta^{(k)} = \alpha s_k + \beta + M \langle \Delta\hat{q}^2 \rangle^{(k-1)}. \tag{7}$$

M, α and β are randomly chosen within ranges that ensure both the input and the feedback contribute comparably to the phase and such to explore a nontrivial region of the observable space.

We begin with a binary task, the temporal XOR, where the target output is $\hat{y}_k = s_k \bigoplus s_{k-1}$ and the input is a random sequence of values $s_k \in \{0, 1\}$. This task requires both nonlinearity and memory of the previous encoded input. The reservoir is trained on a set of inputs by applying linear regression to the measured observables to maximize accuracy (Supplementary Section C).

The performance of the reservoir is then evaluated using the accuracy again, but on a test sequence of new data. In Fig. 3a, we present an example of the test target values and the corresponding experimental prediction. In Fig. 3b,c, we show experimental and simulated accuracies as a function of training set size, for different levels of Gaussian noise. Experimental instabilities can introduce Gaussian noise in the measurement of the observables (Fig. 1), which can be mitigated by averaging over repeated measurements. These instabilities arise from mechanical and thermal fluctuations and are not intrinsic limitations of the protocol, but rather reflect the current performance of the experimental set-up. We see that the training saturates for set sizes around 100 and that noise affects the performance. Nevertheless, the reservoir could achieve experimentally test accuracies of 98 ± 1% for realistic levels of noise, with only 70 training steps. The Digital Twin behaves similarly to the experiment, as shown in Fig. 3c.

A way to further test the memory of the system is through the linear memory task, in which the reservoir is trained to reproduce past entries of the input series. The target function at timestep k is the input encoded τ steps in the past, $\hat{y}_k = s_{k-\tau}$, the input $s_k \in [-1, 1]$ being a random uniformly distributed sequence. Here the three previous observables are not sufficient. Therefore, we resort to spatial multiplexing, which is equivalent to use R different reservoirs at the same time. This is a classical reservoir computing technique used to enhance the expressivity of the system (see Methods for details), whereas the general encoding introduced in the next section leverages spectral multiplexing, an intrinsic resource of the quantum system. In Fig. 3d, we

https://doi.org/10.1038/s41566-026-01880-9

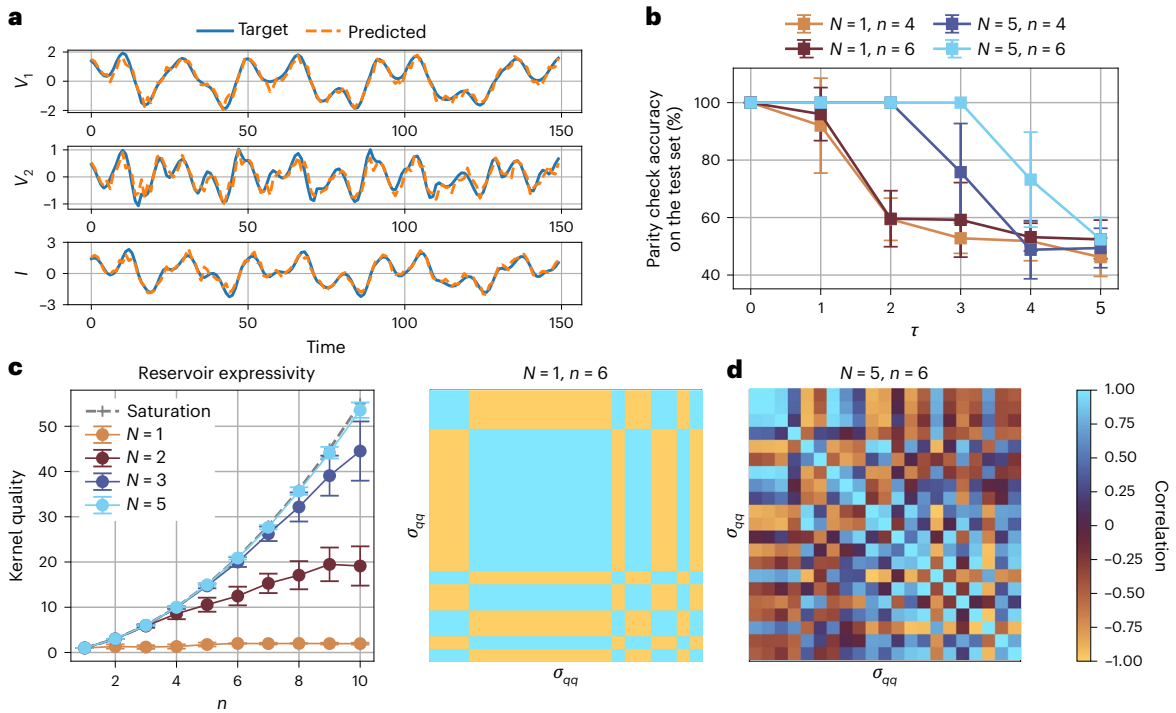


Fig. 4 | Simulated reservoir computing performance: enhanced by general encoding. Double-scroll task (simulated). **a**, Target (blue) and predicted (orange) trajectories using a realistic experimental noise model (Gaussian noise = 0.057), with 15 reservoirs and a training size of 350. Prediction capacities for V_1, V_2 and I are 93.1%, 85.3% and 90.7%, respectively. General encoding (simulated). **b**, Parity check accuracy versus delay τ for different combinations of N (the number of phase segments that we imprint on the pump) and n (the number of measured output modes). Data points represent mean accuracy over $n = 10$ independent trials with different random seeds. Error bars denote ±s.d. across these trials. **c**, Kernel quality (rank of the observables matrix) versus number of measured modes n for different N values, measuring reservoir expressivity. Data points represent mean rank over $n = 25$ independent trials; error bars denote ±s.d. For $N = 1$ (light brown), increasing n does not improve expressivity, while for higher N values, the rank approaches the saturation predicted by a polynomial scaling ($n(n+1)/2$). **d**, Correlation matrices between observables for $N = 1$ and $N = 5$ at $n = 6$ ($n\,\hat{q}$-quadrature operators, yielding $n(n+1)/2 = 21$ observables). For $N = 1$, we can see the high redundancy; for $N = 5$, correlations and thus observables behaviours are richer and more diverse. When the rank of the observables matrix is $n(n+1)/2$, which is its dimension, it means that every observable is meaningful and not redundant.

present an example of target and experimentally predicted values. In Fig. 3e, instead, we show the agreement between the experimental and simulated capacities (see Supplementary Section C for the definition) on test sets when varying τ and the number R of reservoirs. The simulations agree with the experimental results, confirming the advantage of multiplexing (better capacities for higher R) and the fading memory of the system (capacity gradually decreasing for higher delays).

Given the optimal agreement with the experiment, we use the Digital Twin to test the protocol on a more complex task. Concretely, we perform forecasting of chaotic time series such as the double-scroll electronic circuit[49] and also (Supplementary Section H) the Lorenz system. Since the input vectors are three-dimensional, for the global phase encoding, we require at least three spatially multiplexed reservoirs (one per input signal) with additional reservoirs possible to enhance expressivity. The target function is the input one timestep in the future: $\hat{\mathbf{y}}_k = \mathbf{s}_{k+1}$. Concerning the double-scroll, in Fig. 4a, we present an example of target and predicted values, obtained with simulations with realistic noise values. In Supplementary Fig. 2, we present three plots (corresponding to V_1, V_2 and I) where the capacity is calculated while varying the training size and the number of reservoirs R. We prove that even with the experimental noise, we saturate the prediction capacity for relatively small values of R (below 15) and of training size (below 400).

Scaling the expressivity and boosting the memory with general encoding. In the global phase encoding, the expressivity is limited for any number of measured modes $n > 1$ (Methods). Therefore, to take advantage of the multimode nature of the system, we diversify the phase profile of the pump ($N > 1$). This means exploiting fully the phase encoding from equation (4), as shown in Fig. 4. In Fig. 4b, with this general encoding ($N = 5$), we achieve perfect capacities (100%) for a parity check task for $\tau \in \{1, 2, 3\}$ and prove the boost in memory when n, the number of measurement modes, is increased. Furthermore, as shown in Fig. 4c, the expressivity (see Supplementary Section C for definition) increases with n for $N > 1$. Specifically, for $N > 5$, we saturate a polynomial scaling with n as reported in ref. 25: the expressivity scales with $n(n+1)/2$, which is also the dimension of the observables matrix (considering only $\hat{q}$ observables). This means that with a sufficiently large N, all the terms in our covariance matrix are independent. We highlight the increased diversity in behaviours of the observables in Fig. 4d. We also exploit the multidimensionality of the input (it can be a vector of size up to N), to implement again the double-scroll task, without multiplexing. The performances of the global phase encoding with multiplexing ($R = 15$ reservoirs) and the general encoding with N, $n = 9, 9$ are similar, but the gain in scalability is clear as with the general encoding use only one system, instead of multiplexing in 15 reservoirs.

Discussion

Starting from a multimode squeezed resource, we develop an optical QRC platform that leverages spectral shaping capabilities in femtosecond optics for both encoding and decoding information. Fading memory is enabled by linear feedback, even when limited to observables measured at the previous timestep. The CV setting is implemented via coherent homodyne detection, exploiting low-order quadrature moments as observables, an approach identified as offering polynomial advantage in QRC.

By tailoring the pump phase in the parametric process using an EOM, we experimentally show online temporal processing with memory, performing both binary tasks in real time and continuous tasks via successful spatial multiplexing. We demonstrate the nonlinear processing capabilities of the quantum reservoir with a temporal XOR task, assess its memory capacity using continuous input signals and evaluate its forecasting performance with the double-scroll benchmark. Results are reproduced by a Digital Twin, a detailed numerical model of the protocol that includes the properties of the nonlinear optical material, the optical processes, multimode and mode-selective detection schemes, and experimentally derived noise contributions. The Digital Twin enables exploration of the system's capabilities with high fidelity.

The potential of the system's multimode nature is further revealed by encoding information onto a richer phase profile of the pump. This allows consistent nonlinear processing, improved memory (as evidenced by a parity check task) and accurate forecasting of chaotic time series, confirming the advantages of utilizing the full multimode structure. Compared with a purely spatial multiplexing strategy, this approach shows superior scaling, revealing the potential of the system's entangled multimode structure for enhanced performance.

We also provide, in Supplementary Section J, a detailed comparison with classical machine learning approaches (echo state networks and long short-term memory networks) and with another QRC model. Our approach compares favourably with state-of-the-art time-series processing techniques, while relying on inexpensive training (linear regression), requiring only short training sequences and exploiting polynomial scaling between the number of measured modes and the available observables.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41566-026-01880-9.

References

1. Lukoševičius, M. & Jaeger, H. Reservoir computing approaches to recurrent neural network training. *Comput. Sci. Rev.* **3**, 127–149 (2009).
2. Nakajima, K. & Fischer, I. *Reservoir Computing: Theory, Physical Implementations, and Applications* (Springer Nature, 2021).
3. Butcher, J. B., Verstraeten, D., Schrauwen, B., Day, C. R. & Haycock, P. W. Reservoir computing and extreme learning machines for non-linear time-series data analysis. *Neural Netw.* **38**, 76–89 (2013).
4. Ortín, S. et al. A unified framework for reservoir computing and extreme learning machines based on a single time-delayed neuron. *Sci. Rep.* **5**, 14945 (2015).
5. Mujal, P. et al. Opportunities in quantum reservoir computing and extreme learning machines. *Adv. Quantum Technol.* **4**, 2100027 (2021).
6. Ghosh, S., Nakajima, K., Krisnanda, T., Fujii, K. & Liew, T. C. H. Quantum neuromorphic computing with reservoir computing networks. *Adv. Quantum Technol.* **4**, 2100053 (2021).
7. Labay-Mora, A., García-Beni, J., Giorgi, G. L., Soriano, M. C. & Zambrini, R. Neural networks with quantum states of light. *Phil. Trans. A* **382**, 20230346 (2024).
8. Marković, D. & Grollier, J. Quantum neuromorphic computing. *Appl. Phys. Lett.* **117**, 150501 (2020).
9. Fujii, K. & Nakajima, K. Harnessing disordered-ensemble quantum dynamics for machine learning. *Phys. Rev. Appl.* **8**, 024030 (2017).
10. Senanian, A. et al. Microwave signal processing using an analog quantum reservoir computer. *Nat. Commun.* **15**, 7490 (2024).
11. Nokkala, J., Giorgi, G. L. & Zambrini, R. Retrieving past quantum features with deep hybrid classical-quantum reservoir computing. *Mach. Learn. Sci. Technol.* **3**, 035022 (2024).
12. Innocenti, L. et al. Potential and limitations of quantum extreme learning machines. *Commun. Phys.* **6**, 118 (2023).
13. García-Beni, J. et al. Quantum machine learning via continuous-variable cluster states and teleportation. *EPJ Quantum Technol.* **12**, 63 (2025).
14. Kornjača, M. et al. Large-scale quantum reservoir learning with an analog quantum computer. Preprint at https://arxiv.org/abs/2407.02553 (2024).
15. Cimini, V. et al. Large-scale quantum reservoir computing using a Gaussian boson sampler. Preprint at https://arxiv.org/abs/2505.13695 (2025).
16. Mujal, P., Martínez-Peña, R., Giorgi, G. L., Soriano, M. C. & Zambrini, R. Time-series quantum reservoir computing with weak and projective measurements. *npj Quantum Inf.* **9**, 16 (2023).
17. Sannia, A., Giorgi, G. L. & Zambrini, R. Exponential concentration and symmetries in quantum reservoir computing. Preprint at https://arxiv.org/abs/2505.10062 (2025).
18. Shastri, B. J. et al. Photonics for artificial intelligence and neuromorphic computing. *Nat. Photon.* **15**, 102–114 (2021).
19. McMahon, P. L. The physics of optical computing. *Nat. Rev. Phys.* **5**, 717–734 (2023).
20. García-Beni, J., Giorgi, G. L., Soriano, M. C. & Zambrini, R. Scalable photonic platform for real-time quantum reservoir computing. *Phys. Rev. Appl.* **20**, 014051 (2023).
21. Yin, Z. et al. Experimental quantum-enhanced kernel-based machine learning on a photonic processor. *Nat. Photon.* **19**, 1020–1027 (2025).
22. Hoch, F. et al. Quantum machine learning with adaptive boson sampling via post-selection. *Nat. Commun.* **16**, 902 (2025).
23. Sakurai, A., Hayashi, A., Munro, W. J. & Nemoto, K. Quantum optical reservoir computing powered by boson sampling. *Opt. Quantum* **3**, 238–245 (2025).
24. Nerenberg, S., Neill, O. D., Marcucci, G. & Faccio, D. Photon number-resolving quantum reservoir computing. *Opt. Quantum* **3**, 201–210 (2025).
25. Nokkala, J. et al. Gaussian states of continuous-variable quantum systems provide universal and versatile reservoir computing. *Commun. Phys.* **4**, 53 (2021).
26. Suprano, A. et al. Experimental property reconstruction in a photonic quantum extreme learning machine. *Phys. Rev. Lett.* **132**, 160802 (2024).
27. Zia, D. et al. Quantum reservoir computing for photonic entanglement witnessing. *Sci. Adv.* **2025**, eady7987 (2025).
28. Wang, L., Sun, P., Kong, L.-J., Sun, Y. & Zhang, X. Quantum next-generation reservoir computing and its quantum optical implementation. *Phys. Rev. A* **111**, 022609 (2025).
29. Kobayashi, K., Fujii, K. & Yamamoto, N. Feedback-driven quantum reservoir computing for time-series analysis. *PRX Quantum* **5**, 040325 (2024).
30. Monomi, T., Setoyama, W. & Hasegawa, Y. Feedback-enhanced quantum reservoir computing with weak measurements. Preprint at https://arxiv.org/abs/2503.17939 (2025).
31. Hu, F. et al. Overcoming the coherence time barrier in quantum machine learning on temporal data. *Nat. Commun.* **15**, 7491 (2024).
32. Selimović, M. et al. Experimental neuromorphic computing based on quantum memristor. Preprint at https://arxiv.org/abs/2504.18694 (2025).
33. Kouadou, T. et al. Spectrally shaped and pulse-by-pulse multiplexed multimode squeezed states of light. *APL Photon.* **8**, 086113 (2023).
34. Roman-Rodriguez, V. et al. Multimode squeezed state for reconfigurable quantum networks at telecommunication wavelengths. *Phys. Rev. Res.* **6**, 043113 (2024).

35. Renault, P. et al. Experimental optical simulator of reconfigurable and complex quantum environment. *PRX Quantum* **4**, 040310 (2023).
36. García-Beni, J., Giorgi, G. L., Soriano, M. C. & Zambrini, R. Squeezing as a resource for time series processing in quantum reservoir computing. *Opt. Express* **32**, 6733–6747 (2024).
37. Henaff, J. et al. Optical phase encoding in a pulsed approach to reservoir computing. *Opt. Lett.* **49**, 2097–2100 (2024).
38. Ra, Y.-S. et al. Non-Gaussian quantum states of a multimode light field. *Nat. Phys.* **16**, 144–147 (2020).
39. Schuld, M. & Killoran, N. Is quantum advantage the right goal for quantum machine learning? *PRX Quantum* **3**, 030101 (2022).
40. Bourassa, J. E. et al. Blueprint for a scalable photonic fault-tolerant quantum computer. *Quantum* **5**, 392 (2021).
41. Eaton, M., González-Arciniegas, C., Alexander, R. N., Menicucci, N. C. & Pfister, O. Measurement-based generation and preservation of cat and grid states within a continuous-variable cluster state. *Quantum* **6**, 769 (2022).
42. Winnel, M. S., Guanzon, J. J., Singh, D. & Ralph, T. C. Deterministic preparation of optical squeezed cat and Gottesman-Kitaev-Preskill states. *Phys. Rev. Lett.* **132**, 230602 (2024).
43. Anteneh, A., Brunel, L. & Pfister, O. Machine learning for efficient generation of universal photonic quantum computing resources. *Opt. Quantum* **2**, 296–302 (2024).
44. Larsen, M. V. et al. Integrated photonic source of Gottesman–Kitaev–Preskill qubits. *Nature* **642**, 587–591 (2025).
45. Brecht, B., Reddy, D. V., Silberhorn, C. & Raymer, M. G. Photon temporal modes: a complete framework for quantum information science. *Phys. Rev. X* **5**, 041017 (2015).
46. Roman-Rodriguez, V. et al. Continuous variable multimode quantum states via symmetric group velocity matching. *New J. Phys.* **23**, 043012 (2021).
47. Arzani, F., Fabre, C. & Treps, N. Versatile engineering of multimode squeezed states by optimizing the pump spectral profile in spontaneous parametric down-conversion. *Phys. Rev. A* **97**, 033808 (2018).
48. Cai, Y. et al. Multimode entanglement in reconfigurable graph states using optical frequency combs. *Nat. Commun.* **8**, 15645 (2017).
49. Gauthier, D. J., Bollt, E., Griffith, A. & Barbosa, W. A. S. Next generation reservoir computing. *Nat. Commun.* **12**, 5564 (2021).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026, modified publication 2026

Methods

The covariance matrix

The elements of the covariance matrix $\boldsymbol{\sigma}$ are $\sigma_{ij} = 1/2\langle \Delta\hat{x}_i \Delta\hat{x}_j + \Delta\hat{x}_j \Delta\hat{x}_i \rangle$, being $\Delta\hat{x}_i = \hat{x}_i - \langle \hat{x}_i \rangle$ and $\hat{x}_i$ the quadrature $\hat{q}_i$ or $\hat{p}_i$ of the mode i with commutation relation $[\hat{q}_i, \hat{p}_i] = 2\imath$. The covariance matrix is obtained by knowing the interaction Hamiltonian and, in the supermodes basis, takes the form of a diagonal matrix $\sigma(H) = \mathrm{diag}\{e^{2r_1}, e^{2r_2}, ..., e^{-2r_1}, e^{-2r_2}, ...\}$.

Experiment

The experimental set-up is presented in Fig. 1. We use a pulsed laser with a repetition rate of 100 MHz centred at 1,560 nm, and a pulse duration of about 57 fs, which undergoes second harmonic generation in a periodically poled lithium niobate crystal to pump PDC in periodically poled potassium titanyl phosphate waveguides of type 0. More details about the source can be found in ref. 34. In the set-up of this work, an EOM (iXblue NIR-800-LN-0.1) is used to set the phase of the pump beam for the global phase encoding. Phase lock is provided to encode information and measure the desired quadratures at the corresponding phases. The general multispectral phase encoding can be set via a pulse-shaping technique based on a spatial light modulator, as it is implemented in the homodyne set-up to select the spectral basis of the measurement. The average value of squeezing measured in this set-up is −0.45 dB, a reduced value compared with the one in ref. 34 owing to the dispersion introduced in the pump pulse by the fibre-based phase modulator. Such dispersion can be notably reduced in future experiments by shortening the input and output fibre lengths. The anti-squeezing is around 0.7 dB because of losses.

Features and limits of global phase encoding

In experimental control of memory via global phase δ, only three observables that express different dependencies are used. For linear regressions indeed, observables that are correlated (linearly dependent) are redundant[50,51]. For example, measuring $\langle \Delta(\frac{\hat{q}+\hat{p}}{\sqrt{2}})^2 \rangle$ is equivalent to measuring $\langle \Delta\hat{q}\Delta\hat{p} \rangle$ since analytically, for a global phase encoding, $\langle \Delta\hat{q}\Delta\hat{p} \rangle(\delta) = \langle \Delta(\frac{\hat{q}+\hat{p}}{\sqrt{2}})^2 \rangle(\delta) + cst$ (as shown in Supplementary Section D).

Moreover, when encoding on a global phase δ, even if we increase the number of measured modes $n > 1$, all the terms of the covariance matrix behave like either $\sin^2\delta$, $\cos^2\delta$ or $\cos\delta \sin\delta$. The expressivity of the reservoir is therefore limited for all n. This also happens when simulating the system with a detailed physical model that takes into account the characteristics of the PDC material as shown in the bottom part of Fig. 4, where we encode on a single pump band ($N = 1$) and measure on n output frequency bands (frexels). Indeed, to evaluate more quantitatively the expressivity of the system, we compute the rank of a matrix whose rows correspond to observables evaluated at different input states. This rank, called kernel quality in the literature[50–52], would indicate the number of linearly independent observables; however, since we scaled them beforehand, it also takes into account affine dependence. For linear regressions, an observable affine to another should not be useful. Since for a global phase, the observables behave like $\sin^2\delta = 1 - \cos^2\delta$, $\cos^2\delta$ or $\cos\delta \sin\delta$, analytically the rank should be equal to 2 for all n; however, numerical instabilities of the Schmidt decomposition yield small fluctuations. For example, we can see that there are two behaviours only ($O_A \sim \cos^2\delta$ and $O_B \sim -\cos^2\delta$) from the first subplot in Fig. 4d, where we consider correlations in $\hat{q}$ quadratures alone.

Simulations with the Digital Twin

Our experiment is fully characterized and accurately reproducible via simulation. This allows for faithful reproduction of experimental results and reliable predictions for new configurations, including the impact of additions such as a spatial light modulator on the pump. Although simulations are much slower than the femtosecond-scale nonlinear processes in the waveguides, they constitute a useful Digital Twin of the experiment, enabling interpretation and hypothesis testing.

We begin by modelling the nonlinear PDC process in the waveguides. This involves computing the JSA, determined by the phase-matching function, derived from the waveguide geometry and material (via the Sellmeier equations), and by the complex pump beam profile, which in our protocols encodes the input information. From the JSA, we extract the Schmidt modes and squeezing coefficients via singular value decomposition, the most computationally intensive step. We model the measurement stage by computing overlaps between Schmidt and measurement modes, which allows us to finally reconstruct expected observables for arbitrary pump phase profiles. The core of the simulations stems from a detailed physical model shown in ref. 46 and implemented in a Python code (all the results can be retrieved in ref. 53).

Spatial multiplexing

At each timestep, the same input s_k is fed to all R reservoirs in parallel, each with their different parameters α_r and β_r ($r \in 1, ..., R$); this ensures that the observables of the R reservoirs are not correlated. Concerning the feedback term, we collect the R observables $\langle \Delta\hat{q}^2_{(r)} \rangle^{(k-1)}$ and we distribute them in the R inputs with a mask M of size $R \times R$. Each reservoir has then the following input at timestep k:

$$\delta_r^{(k)} = \alpha_r s_k + \beta_r + \sum_{r'=1}^{R} M_{r,r'} \langle \Delta\hat{q}^2_{(r')} \rangle^{(k-1)} \quad r \in 1, \ldots, R. \tag{8}$$

The mask is a full matrix, meaning that the feedback term of each reservoir r carries the observable values from all R reservoirs. Experimentally, the parallelization of the process was implemented sequentially, sending the same input R times to the physical set-up but with different parameters for every repetition and collecting the observables to constitute the feedback term at the end of the cycle.

Pathway towards non-Gaussian features

While polynomial scaling is achievable with Gaussian states, surpassing this performance necessitates the inclusion of non-Gaussian elements. Implementable candidates include spectral mode-selective photon subtraction, already demonstrated for spectrally multimode light fields[38]. Resources with stronger non-Gaussianity can be introduced via photon-number-resolving detection in similar spectral mode-selective processes. In fact, photon-number-resolving detectors have been used in integrated experimental set-ups to generate strongly non-Gaussian states such as Gottesman–Kitaev–Preskill states[44]. They have also been shown to enable nearly deterministic schemes for the generation of Gottesman–Kitaev–Preskill[43] and Schrödinger cat states[42]. The experimental effort should go hand in hand with theoretical investigations of the advantages provided by non-Gaussianity in the CV setting, that is, based on homodyne data readout. We can reasonably expect that even weak non-Gaussianity, introduced by heralding on a low number of detected photons, would enhance the effective nonlinearity of the platform.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The experimental datasets generated and analysed during the current study are stored in a laboratory repository and are available from the corresponding authors on reasonable request owing to the large sizes. All of the simulated data that support the findings of this study are publicly available via GitHub at https://github.com/EQ15T/optical_qrc (ref. 53).

Code availability

The code that supports the findings of this study is publicly available via GitHub at https://github.com/EQ15T/optical_qrc (ref. 53).

References

50. Appeltant, L. et al. *Reservoir Computing Based on Delay-Dynamical Systems*. These de Doctorat, Vrije Universiteit Brussel/Universitat de les Illes Balears (2012).
51. Wijesinghe, P., Srinivasan, G., Panda, P. & Roy, K. Analysis of liquid ensembles for enhancing the performance and accuracy of liquid state machines. *Front. Neurosci.* **13**, 504 (2019).
52. Wringe, C., Trefzer, M. & Stepney, S. Reservoir computing benchmarks: a tutorial review and critique. *Int. J. Parallel Emergent Distrib. Syst.* **40**, 313–351 (2025).
53. Paparelle, I. et al. EQ15T/optical_qrc. *GitHub* https://github.com/EQ15T/optical_qrc (2025).

Acknowledgements

This work was supported by the European Research Council under the Consolidator Grant COQCOoN (grant number 820079) (V.P.) and by Agence Nationale de la Recherche (OQuLus, ANR-22-PETQ-0013) (V.P.). We acknowledge the Spanish State Research Agency, through the María de Maeztu project CEX2021-001164-M (R.Z.), through the COQUSY project PID2022-140506NB-C21 and -C22 (G.L.G. and R.Z.), through the INFOLANET project PID2022-139409NB-I00 (M.C.S.) and through the QuantCom project CNS2024-154720 (G.L.G.), all funded by MICIU/AEI/10.13039/50110001103 (the PIDs also by ERDF, EU); the project is funded under the Quantera II programme that has received funding from the EU's H2020 research and innovation programme under the GA number 101017733, and from the Spanish State Research Agency, PCI2024-153410 funded by MICIU/AEI/10.13039/50110001103 and cofunded by the European Union (G.L.G.); MINECO through the QUANTUM SPAIN project (R.Z.), and EU through the RTRP—NextGenerationEU within the framework of the Digital Spain 2025 Agenda; CSIC's Quantum Technologies Platform (QTEP). J.G.-B. is funded by the Conselleria d'Educació, Universitat i Recerca of the Government of the Balearic Islands with grant code FPI/036/2020 and D.M. by the 'la Caixa' Foundation (ID 100010434), code LCF/BQ/DR24/12080033.

Author contributions

I.P., J.H., G.L.G., M.C.S., R.Z. and V.P. conceived and designed the experiments. I.P. and J.H. performed the experiments. I.P., E.G. and D.M. analysed the data. I.P., J.G.-B., E.G., D.M., G.L.G., M.C.S., R.Z. and V.P. contributed materials and analysis tools. I.P., D.M., R.Z. and V.P. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41566-026-01880-9.

Correspondence and requests for materials should be addressed to Iris Paparelle or Valentina Parigi.

Peer review information *Nature Photonics* thanks Xiangdong Zhang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

nature portfolio

Corresponding author(s): Iris Paparelle

Last updated by author(s): Feb 19, 2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data were acquired using an oscilloscope and a spectrum analyzer interfaced via custom Python scripts written in-house. Instrument communication was handled using PyVISA and vendor-specific drivers (TeledyneLeCroyPy), and hardware control included the Pyrpl library. Data processing was performed using Python with NumPy, SciPy, Pandas, and Matplotlib. No commercial acquisition software was used. The custom code used in this study is available from the corresponding author upon reasonable request. Regarding simulations, all data is available on https://github.com/EQ15T/optical_qrc.
Data analysis	Data processing and analysis was performed using Python with NumPy, SciPy, Pandas, and Matplotlib. No commercial acquisition software was used. The custom code used in this study is available from the corresponding author upon reasonable request. Regarding simulations, all data is available on https://github.com/EQ15T/optical_qrc.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The experimental datasets generated and analysed during the current study are stored in a laboratory repository and are available from the corresponding author on reasonable request due to their large size. All code and simulated data supporting the findings of this study are publicly available at the repository described in https://github.com/EQ15T/optical_qrc.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Reporting on race, ethnicity, or other socially relevant groupings	*Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status). Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.) Please provide details about how you controlled for confounding variables in your analyses.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	*Describe how sample size was determined, detailing any statistical methods used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data exclusions	*Describe any data exclusions. If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Replication	*Describe the measures taken to verify the reproducibility of the experimental findings. If all attempts at replication were successful, confirm this OR if there are any findings that were not replicated or cannot be reproduced, note this and describe why.*
Randomization	*Describe how samples/organisms/participants were allocated into experimental groups. If allocation was not random, describe how covariates were controlled OR if this is not relevant to your study, explain why.*

Blinding	*Describe whether the investigators were blinded to group allocation during data collection and/or analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional, quantitative experimental, mixed-methods case study).*
Research sample	*State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.*
Sampling strategy	*Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.*
Data collection	*Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.*
Timing	*Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Non-participation	*State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.*
Randomization	*If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.*

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.*
Research sample	*Describe the research sample (e.g. a group of tagged Passer domesticus, all Stenocereus thurberi within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.*
Sampling strategy	*Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data collection	*Describe the data collection procedure, including who recorded the data and how.*
Timing and spatial scale	*Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Reproducibility	*Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.*
Randomization	*Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.*
Blinding	*Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	*Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).*
Location	*State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).*
Access & import/export	*Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).*
Disturbance	*Describe any disturbance caused by the study and how it was minimized.*

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☐ Antibodies
☐	☐ Eukaryotic cell lines
☐	☐ Palaeontology and archaeology
☐	☐ Animals and other organisms
☐	☐ Clinical data
☐	☐ Dual use research of concern
☐	☐ Plants

Methods

n/a	Involved in the study
☐	☐ ChIP-seq
☐	☐ Flow cytometry
☐	☐ MRI-based neuroimaging

Antibodies

Antibodies used	*Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.*
Validation	*Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.*

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	*State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.*
Authentication	*Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.*
Mycoplasma contamination	*Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.*
Commonly misidentified lines (See ICLAC register)	*Name any commonly misidentified cell lines used in the study and provide a rationale for their use.*

Palaeontology and Archaeology

Specimen provenance	*Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.*
Specimen deposition	*Indicate where the specimens have been deposited to permit free access by other researchers.*
Dating methods	*If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where*

Dating methods	*they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.*

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight	*Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	*For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.*
Wild animals	*Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.*
Reporting on sex	*Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.*
Field-collected samples	*For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.*
Ethics oversight	*Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	*Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.*
Study protocol	*Note where the full trial protocol can be accessed OR if not available, explain why.*
Data collection	*Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.*
Outcomes	*Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.*

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
☐	☐	Public health
☐	☐	National security
☐	☐	Crops and/or livestock
☐	☐	Ecosystems
☐	☐	Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes	
☐	☐	Demonstrate how to render a vaccine ineffective
☐	☐	Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐	Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐	Increase transmissibility of a pathogen
☐	☐	Alter the host range of a pathogen
☐	☐	Enable evasion of diagnostic/detection modalities
☐	☐	Enable the weaponization of a biological agent or toxin
☐	☐	Any other potentially harmful combination of experiments and agents

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.

ChIP-seq

Data deposition

☐ Confirm that both raw and final processed data have been deposited in a public database such as GEO.

☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links
May remain private before publication.

For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

Files in database submission

Provide a list of all files available in the database submission.

Genome browser session
(e.g. UCSC)

Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates

Describe the experimental replicates, specifying number, type and replicate agreement.

Sequencing depth

Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.

Antibodies

Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.

Peak calling parameters

Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.

Data quality

Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.

Software

Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.

Flow Cytometry

Plots

Confirm that:

☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☐ All plots are contour plots with outliers or pseudocolor plots.

☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	*Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.*
Instrument	*Identify the instrument used for data collection, specifying make and model number.*
Software	*Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.*
Cell population abundance	*Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.*
Gating strategy	*Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.*

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type	*Indicate task or resting state; event-related or block design.*
Design specifications	*Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.*
Behavioral performance measures	*State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).*

Acquisition

Imaging type(s)	*Specify: functional, structural, diffusion, perfusion.*
Field strength	*Specify in Tesla*
Sequence & imaging parameters	*Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.*
Area of acquisition	*State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.*

Diffusion MRI ☐ Used ☐ Not used

Preprocessing

Preprocessing software	*Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).*
Normalization	*If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.*
Normalization template	*Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.*
Noise and artifact removal	*Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).*

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See Eklund et al. 2016)

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.